



Cross-phylum extrapolation of the *Daphnia magna* chronic biotic ligand model for zinc to the snail *Lymnaea stagnalis* and the rotifer *Brachionus calyciflorus*

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ABSTRACT

We investigated if the chronic zinc biotic ligand model (BLM) developed earlier for the arthropod *Daphnia magna* could be extrapolated to predict chronic ecotoxicity of zinc as a function of water chemistry to two species from other phyla, i.e. the mollusc *Lymnaea stagnalis* and the rotifer *Brachionus calyciflorus*. We chronically exposed these two species to zinc in six natural surface waters. These water covered a wide range of pH (6.8–8.3), dissolved organic carbon (1.2–12.7 mg/L) and Ca (8.8–118 mg/L). Across all waters tested, the 28d-EC10s (200–1629 µg Zn/L) and EC50s (382–2026 µg Zn/L) for *L. stagnalis* spanned a 8.1-fold and 5.3-fold range, respectively. The 2d-EC10s (142–550 µg Zn/L) and 2d-EC50s (195–1104 µg Zn/L) for *B. calyciflorus* spanned a 3.9-fold and 5.7-fold range, respectively. The data indicated that higher pH and higher concentrations of Ca and DOC were generally associated with lower toxicity (higher ECx values). Furthermore, the chronic Zn BLM for *D. magna*, when calibrated only to reflect the intrinsic sensitivity of *L. stagnalis* and *B. calyciflorus*, was able to predict all ECx values with a less than 1.6-fold error, which demonstrates that the chronic *D. magna* Zn BLM can be extrapolated to other invertebrate phyla. This lends further support to the use of the chronic Zn BLM to account for bioavailability of zinc in aquatic risk assessment and the derivation of environmental quality standards.

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1. Introduction

Recently, the European Union Risk Assessment Report (EU RAR) on zinc and zinc compounds has been published in the framework of EU Council Regulation 793/93/EEC and this report includes a section on risk assessment for freshwater (ECB, 2008). Because physico-chemical variables like pH, Ca and dissolved organic carbon (DOC) can influence chronic zinc toxicity to freshwater biota, and because these influences can be accurately predicted with chronic biotic ligand models (BLMs) (Nigoyi and Wood, 2004; De Schampelaere et al., 2005; Van Sprang et al., 2009), the EU RAR used these BLMs to implement a method for bioavailability correction of risk estimates in the risk characterisation (Bodar et al., 2005; ECB, 2008). However, field-validated chronic Zn BLMs only existed for three species, i.e. the green alga *Pseudokirchneriella subcapitata* (De Schampelaere et al., 2005); the herbivorous crustacean *Daphnia magna* (Heijerick et al., 2005) and the fish *Oncorhynchus mykiss* (De Schampelaere and Janssen, 2004), and not for each individual species for which chronic zinc toxicity data were available in the compiled aquatic toxicity database (ECB, 2008; Van Sprang et al., 2009). As a consequence, a relatively pragmatic and inherently conservative bioavailability correction method, i.e. the “bioavailability factor approach” (BioF

approach), was finally selected. For numerical details of this approach, please consult the Zn RAR document (ECB, 2008) and Van Sprang et al. (2009). Recently, Van Sprang et al. (2009) advocated the use of another approach, called the “full species sensitivity distribution (SSD) bioavailability normalization” method. In brief, this method uses the chronic Zn BLMs for bioavailability normalization of all available high quality chronic ecotoxicity data for all species. With this method region, river or even site-specific physico-chemistry can be used to estimate hazardous concentrations for 5% of the organisms (HC5) by fitting a species sensitivity distribution (SSD) to the entire normalized ecotoxicity dataset (Van Sprang et al., 2009). This method heavily relies on the assumption that chronic Zn BLMs can be extrapolated across species within the same trophic level. In other words, these BLMs should be able not only to accurately predict how differences in physico-chemistry cause differences in observed toxicity to those species for which these BLMs were specifically developed, they should also be able to do so for other species. Substantial evidence already exists that BLMs for Cu and Ni can be extrapolated across species to explain observed toxicity differences among synthetic media and/or field waters with various physico-chemical conditions (Deleebeeck et al., 2007; 2009; De Schampelaere et al., 2006; De Schampelaere and Janssen, 2006; Van Sprang et al., 2008; ECB, 2009). As a consequence the full SSD normalization approach has been adopted in regulatory context for both copper (USEPA, 2007; Van Sprang et al., 2008) and nickel (ECB, 2009). However, evidence supporting such extrapolations with respect to chronic Zn BLMs is still scarce.

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Van Sprang et al. (2009) provided – on the basis of an analysis of existing literature data – support for the extrapolation of Zn BLMs across fish and algae species. However, no such support could be lent to the extrapolation of the chronic Zn BLM developed for *D. magna* toward species from other invertebrate phyla. Therefore, the objective of the present study was to test this empirically. We first investigated the chronic toxicity of Zn to the rotifer *Brachionus calyciflorus* and the pulmonate freshwater snail *Lymnaea stagnalis* in a total of six Zn-spiked field waters with a range of physico-chemical characteristics representative of the variation encountered in EU water bodies. We then tested if the *D. magna* chronic Zn BLM would be able to accurately predict chronic Zn toxicity for *B. calyciflorus* and *L. stagnalis* in these waters, after adjustment for inherent sensitivity.

2. Materials and methods

2.1. Collection of field waters

Samples of natural surface waters were taken from four unpolluted water bodies: the Ankeveensche Plasschen (Nederhorst-Den-Berg, The Netherlands, further denoted 'Ankeveen'), Markermeer (Marken, The Netherlands, further denoted 'Markermeer'), Le Vayon (Trélon, France, further denoted 'Vayon'), and L' Ourthe Orientale (Brisy, Belgium, further denoted 'Brisy'). Fifty liters of Ankeveen, Markermeer, and Vayon water and 150 L of Brisy water were pumped and filtered through 0.45 µm filters in the field and collected in 50 L acid washed (0.14 N HNO₃) polyethylene vessels. Upon arrival in the laboratory (within 4 h of sampling) the sample was immediately stored at 4 °C in the dark until use. Subsamples of these four waters were taken for analysis of Cl[−] and SO₄^{2−} concentrations.

2.2. Preparation of test media

In addition to the four field-collected waters, we prepared two more media by modification of the pH and Ca concentration of the Brisy water, which was naturally low in pH (6.8) and Ca (10 mg/L). The rationale of this was that this would allow us to investigate the individual effect of pH and Ca on chronic Zn toxicity to *L. stagnalis* and *B. calyciflorus*. The first modified water, further denoted 'Brisy, high pH' was obtained by adding 3 mmol/L of NaHCO₃ to 100 L of Brisy water. Following equilibration with ambient atmospheric CO₂ this yielded a pH of about 8.2. The second modified water, further denoted 'Brisy, high pH, high Ca', was obtained by adding 2.5 mmol CaCl₂ to 50 L of the 'Brisy, high' pH medium.

Thus, a total of six test media were available, which were then used as the dilution water to prepare a zinc concentration series for ecotoxicity testing, which included a negative control (no Zn added). A Zn stock solution was prepared by dissolution of ZnCl₂ in deionized water (>99.9% purity) and this stock solution was used to prepare all Zn treatments. The nominal zinc concentrations added to all media is given as [supportive information](#). For the 28-day exposures with snails, fresh zinc spiked test media were prepared weekly, while for the 2-day exposures with rotifers, fresh zinc spiked media were prepared one day before test initiation.

2.3. Ecotoxicity tests with *L. stagnalis*

Fourteen-day old *L. stagnalis* (Linnaeus, 1758) snails were kindly provided by Dr. Joris Koene from the Vrije Universiteit in Amsterdam, where a continuous culture of these animals is maintained in city tap water (pH 8.4, hardness 150 mg CaCO₃/L) at 20 °C. Upon arrival in our laboratory, 400 snails were acclimated to each of six test media (without added Zn) in polyethylene aquaria holding 3 L of medium for 1 week prior to initiating exposure to Zn. During the acclimation, snails were fed *ad libitum* with lettuce and transferred to fresh medium every other day. Ecotoxicity tests in all six test media were run simultaneously to

minimize temporal variability. At test initiation, snails were 21 days old and their shell length was on average 7.08 mm. All snails were housed individually in a polyethylene test container holding 200 mL of test medium. Eight snails were exposed to the control and to each Zn treatment. The tests were conducted at 20 °C under a light-cycle of 12 h light – 12 h dark. For all treatments, the entire 200 mL of 'old' test medium was replaced with fresh test medium twice a week (on the 4th, 7th, 11th, 14th, 18th, 21st, and 25th day of exposure). Before renewal, fresh solutions were adjusted to the target pH using dilute HCl or NaOH. pH was also measured in old solutions at every renewal. On several occasions (see [supportive information](#) for details), samples were taken from fresh and old (just before renewal) exposure solutions for analysis of total Zn, dissolved Zn, DOC and dissolved inorganic carbon (DIC). Dissolved samples were obtained by filtering through 0.45 µm filters (Acrodisc, PALL Life Sciences, Port Washington, NY, USA). Samples for major cations (Ca, Mg, Na, K) were taken at test initiation. Samples for Zn and major cations were acidified by adding ultrapure HNO₃ (Normatom quality, VWR Prolabo, Leuven, Belgium) to a final concentration of 0.14 mol/L. At test initiation and with every medium renewal, each snail was provided with an *ad libitum* food ration of 40, 60, 80, and 100 mg lettuce in the first, second, third and fourth week of the exposure, respectively. During the 28-day experiment, mortality was scored daily and the shell length of each surviving individual was measured to the nearest 10 µm at test initiation and at days 7, 14, 21 and 28 of the exposure. The exponential length-based growth rate (µ) of each individual snail was calculated as the slope of the linear regression of log_e transformed length readings versus day of exposure (5 data points per snail). Only growth rates of snails that survived the full 28 days of exposure were considered for fitting concentration response curves (see further).

2.4. Ecotoxicity tests with *B. calyciflorus*

Ecotoxicity tests with *B. calyciflorus* were conducted using the ROTOXKIT F following the standard operational procedure (SOP) (Microbiotests, Gent, Belgium), which closely follows the protocol No. 8420 of the American Public Health Association (Clesceri et al., 1998) for the 48-hour life-cycle test with *B. calyciflorus*. Juveniles were hatched from dormant eggs following the ROTOXKIT F SOP and <two-hour hatchlings were immediately used in the toxicity experiments with all six media. As for the snail tests, tests in all media were run simultaneously. Toxicity experiments were performed in polyethylene 24-multiwell plates with wells containing 1 mL of test solution. Algal food was added to each well to a concentration of 2 × 10⁶ cells/mL. After addition of food, all test cups were inoculated with one rotifer hatchling, with 10 replicates per Zn concentration. The multi-wells were then closed and incubated for 48 h at 25 °C in darkness. At the end of the incubation the number of live rotifers was counted in each cup with the aid of a dissecting microscope. The intrinsic rate of increase (r, d^{−1}) of asexually reproducing rotifers was used as the toxicity endpoint and was calculated as log_e (number of live rotifers at end of test) / 2 (Clesceri et al., 1998; De Schamphelaere et al., 2006).

pH was measured at test initiation and test termination. Furthermore, samples from exposure solutions were taken at test initiation and termination for analysis of total Zn, dissolved Zn, DOC and dissolved inorganic carbon (DIC). Dissolved samples were obtained by filtering through 0.45 µm filters (Acrodisc, PALL Life Sciences, Port Washington, NY, USA). Dissolved samples (0.45 µm) for major cations (Ca, Mg, Na, K) were taken at test initiation. Samples for Zn and major cations were acidified by adding ultrapure HNO₃ (Normatom quality, VWR Prolabo, Leuven, Belgium) to a final concentration of 0.14 mol/L.

2.5. Analytical chemistry

All analyses included the use of appropriate blanks and calibration standards. In snail and rotifer tests, pH was measured with glass

Table 1

Stability constants of the chronic Zn biotic ligand model (BLM) for *Daphnia magna* (Heijerick et al., 2005) and intrinsic sensitivities ($f_{\text{ZnBL,critical}}$) of several species expressed as critical fractions of the biotic ligand occupied by Zn at a low chronic effect level.

Stability constant ^a	Value	Species	$f_{\text{ZnBL,critical}}$
Log K_{CaBL}	3.2	<i>C. dubia</i>	0.0127 ^b
Log K_{MgBL}	2.7	<i>P. jenkinsii</i>	0.0130 ^b
Log K_{NaBL}	1.9	<i>H. azteca</i>	0.0147 ^b
Log K_{HBL}	5.8	<i>A. fissa</i>	0.0480 ^b
Log K_{ZnBL}	5.3	<i>B. rubens</i>	0.0480 ^b
		<i>D. magna</i>	0.0729 ^b
		<i>D. polymorpha</i>	0.0785 ^b
		<i>D. longispina</i>	0.0977 ^b
		<i>B. calyciflorus</i>	0.121 ^c
		<i>E. virgo</i>	0.194 ^b
		<i>L. stagnalis</i>	0.203 ^c

^a Biotic Ligand Model stability constants for XBL are defined as $K_{\text{XBL}} = [\text{XBL}] / \{(\text{X}) \times [\text{BL}]\}$, where parentheses (X) denotes chemical activity of X, [XBL] is the concentration of X bound to the BL, and [BL] is the concentration of unoccupied BL sites.

^b Calculated by Van Sprang et al. (2009).

^c This study (see Table 3).

electrodes from Hanna Instruments (HI1618D, Woonsocket, RI, USA). Zn concentrations were measured using flame atomic absorption spectrophotometry (SpectraAA100, Varian, Mulgrave, Australia). The detection limit for Zn was 5 µg/L and measured Zn concentrations in all negative control test waters were below this detection limit. Two certified reference samples, TMDA-62 and TM-25.2 (National Water Research Institute, Burlington, ON, Canada) with certified Zn concentrations (mean ± 95% confidence interval) of $110 \pm 15.5 \mu\text{g L}^{-1}$ and $24 \pm 4.6 \mu\text{g L}^{-1}$, respectively, were analyzed at the beginning and end of each series of Zn measurements. Measured values were always within 10% of the certified value. Dissolved organic carbon (DOC) and dissolved inorganic carbon (DIC) were measured with a Total Organic Carbon analyzer (TOC-5000, Shimadzu, Duisburg, Germany). DOC was measured in non-purgeable organic carbon (NPOC) mode. The macro-anions Cl^- and SO_4^{2-} were measured with colorimetric SpectroQuant® kits (Merck, Darmstadt, Germany) using an Aquamate spectrophotometer (Thermo Electron Corporation; Cambridge, UK). The macro-cations Ca, Mg, Na and K were measured using a flame emission spectrophotometer (ELEX 6361, Eppendorf, Hamburg, Germany).

2.6. Concentration response models

The raw concentration response data, i.e. measured dissolved Zn concentrations and growth rates for each replicate of each concen-

tration were inserted in Statistica 6.0 software (Statsoft, Tulsa, OK, USA) for calculating 10% (EC10) and 50% effective concentrations (EC50). For *L. stagnalis* and *B. calyciflorus* 28d-ECx and 2d-ECx values were calculated, respectively. The dissolved Zn concentration used for calculations was the arithmetic mean of (1) the arithmetic mean of all dissolved Zn measurements in fresh solutions and (2) the arithmetic mean of all dissolved Zn measurements in old solutions. Unless noted otherwise, a three-parameter log-logistic concentration response model was fitted to the data (De Schamphelaere and Janssen, 2004):

$$\mu \text{ or } r = \frac{k}{1 + \left(\frac{[\text{Zn}]_{\text{dissolved}}}{\exp(a)} \right)^{\frac{\ln(\frac{1}{b})}{b-a}}} \quad (3)$$

where μ or r = the predicted shell length growth rate (snails) or intrinsic rate of increase (rotifers) (d^{-1}), $[\text{Zn}]_{\text{dissolved}}$ = the dissolved Zn concentration (µg/L), k = the control response (d^{-1}), $a = \log_e$ (EC50 as µg/L), and $b = \log_e$ (EC10 as µg/L). Parameter estimation and calculation of the 95% confidence limits were carried out using the Levenberg–Marquardt algorithm as implemented in Statistica.

2.7. BLM modeling

All BLM modeling was performed with BLM software (version 2.1.2 for Windows®, Hydroqual, Mahwah, NJ, USA), which incorporates WHAM-Model V (Tipping, 1994) as the geochemical speciation module. Stability constants for the formation of inorganic Zn-complexes were taken from the National Institute for Standardization (NIST) (Smith et al., 2004). We assumed that natural organic matter (NOM) consisted of 61% active fulvic acid and 39% material inert for Zn binding, and that the elemental C content of NOM is 50% on a mass basis (see Cheng et al., 2005). These assumptions are realized by setting the $\text{DOC}_{\text{input}}$ concentration in the BLM software at $0.61 \times$ the measured DOC concentration and the % humic acid to 0.01% (De Schamphelaere et al., 2005). The stability constants used were those reported for the chronic Zn BLM for *D. magna* (Table 1, Heijerick et al., 2005). The thermodynamics database file used (water12.dbs) and the BLM parameter file (Zn_Daphnia_magna_chronic_21d_EC50.dat), which holds the stability constants of the chronic Zn BLM from Table 1 and which is needed to run the BLM software, are given as supportive information (S1 and S2, respectively). The DOC, pH (both reported in Table 2) and DIC values (see supportive information) used for BLM modeling are the means of fresh and old solutions of those Zn treatments encompassing EC10 and EC50 values (for calculations from raw data, see supportive information S3 for *L. stagnalis* and S4 for

Table 2

Main physico-chemical properties of the six exposure media and the 10% (EC10) and 50% effective concentration (EC50) of dissolved Zn to *Lymnaea stagnalis* (28 days) and *Brachionus calyciflorus* (2 days), with their 95% confidence interval (CI).

Medium	pH	DOC (mg/L)	Ca (mg/L)	Mg (mg/L)	Na (mg/L)	EC10 (µg/L)	95% CI	EC50 (µg/L)	95% CI
<i>Lymnaea stagnalis</i>									
Ankeveen	7.81	12.7	79.4	14.0	64.8	1629	(1034–2566)	2026	(1910–2148)
Markermeer	7.90	7.82	64.4	15.7	75.6	910	(763–1086)	1284	(1176–1403)
Voyon	7.41	2.85	8.8	3.8	7.1	200	(139–289)	382	(336–434)
Brisy	6.77	1.50	9.8	4.0	8.0	244	(195–306)	407	(367–452)
Brisy high pH	8.28	1.71	9.7	3.9	75.4	330	(252–431)	568	(501–643)
Brisy high pH, high Ca	8.25	1.53	112.0	3.8	76.0	719	(586–883)	1118	(1002–1247)
<i>Brachionus calyciflorus</i>									
Ankeveen	7.77	8.94	79.2	13.9	64.4	550	(359–844)	1104	(920–1326)
Markermeer	7.86	6.28	65.7	15.7	75.8	ND ^a		>882 ^b	
Voyon	7.40	2.83	11.4	4.2	11.1	197 ^c	(171–226)	244	(214–278)
Brisy	6.89	1.18	10.9	4.8	14.1	142	(122–165)	195	(175–216)
Brisy high pH	8.09	1.73	10.2	3.9	83.9	66	(26–166)	275	(191–397)
Brisy high pH, high Ca	8.16	1.49	118	3.9	87.0	453	(293–703)	721	(621–838)

^a Could not be determined, no reliable response model could be fitted.

^b No reliable response model could be fitted, 882 µg/L resulted in a growth rate reduction of 31%.

^c Hormesis model of Van Ewijk and Hoekstra (1993) was fitted.

B. calyciflorus). Values for all other physico-chemical variables used for modeling are those measured at test initiation (see Table 2 for Ca, see supportive information S3 and S4 for all other variables).

3. Results and discussion

3.1. Physico-chemistry of test media

pH and concentrations of DOC and Ca are the three variables that have the largest impact on chronic Zn bioavailability (De Schamphelaere et al., 2005; Van Sprang et al., 2009). The investigated range (Table 2) of pH (6.8–8.3), DOC (1.2–12.7 mg/L) and Ca (8.8–118.0 mg/L) covers most of the 10th to 90th percentile range of the FOREGS database (www.gsf.fi/foregs/geochem, Salminen et al., 2005) and more than the 10th to 90th percentile range of the GEMS database (www.gemswater.org; Heijerick et al., 2003; ECB, 2008) (Fig. 1). As both databases report stream water chemistry across the entire EU, this indicates that the results obtained with the set of natural and modified natural exposure media investigated here are relevant for interpretation in the context of the EU risk assessment of Zn and any related regulatory frameworks (e.g., REACH and the Water Framework Directive).

3.2. Ecotoxicity test results

Raw concentration response data are given as supportive information (S3 and S4). Summarized concentration response data and fitted concentration response curves are presented in Fig. 2 (*L. stagnalis*) and Fig. 3 (*B. calyciflorus*). The EC10s and EC50s and their 95% CI for both species in all exposure media are given in Table 2. Reliable EC10s and EC50s could be calculated in most cases, with two exceptions for the rotifer experiments. First, in Markermeer water, the rotifers only experienced a significant effect (31% reduction of intrinsic rate of increase) at the highest tested concentration of 882 µg Zn/L (Fig. 3). As such, no reliable concentration response model could be fitted to these data. Second, in Brisy, high pH water, the 95% CI of the 2d-EC10 spanned a 6.3-fold range, making this value much less reliable than all other EC10 and EC50 values derived. Indeed, a maximum 2.5-fold range of the 95% CI was observed for all other EC10s and EC50s for both organisms (Table 2). Therefore, we will not further consider this specific EC10 for parameterization of the BLM or for assessing the overall accuracy of BLM predictions (see 3.4). In one case, i.e. the rotifer test in Voyon water, the log-logistic model could not be fitted to the data (no convergence of the fitting procedure). However, a log-logistic model including a hormesis effect (Van Ewijk and Hoekstra, 1993) provided a reasonable fit. While hormesis, i.e. the stimulation of performance at low concentrations of toxic chemicals, has often been observed in ecotoxicology (Calabrese, 2008), it is unclear why we only observed it in *B. calyciflorus* in one of

the six test waters investigated. It is possible, though, that as an artifact of the relatively high variability of growth rates in some treatments (including the control), the hormetic concentration response model could be fitted while the simple log-logistic model could not.

Overall, the 28d-EC10s (200–1629 µg/L) and EC50s (382–2026 µg/L) for *L. stagnalis* spanned a 8.1-fold and 5.3-fold range, respectively (Table 2). The reliable 2d-EC10s (142–550 µg/L) and 2d-EC50s (195–1104) for *B. calyciflorus* spanned a 3.9-fold and 5.7-fold range, respectively (Table 2). Hence, based on the EC50 estimates, which are more precise than the EC10s, it can be suggested that the variation in chronic Zn toxicity due to differences in water chemistry is similar for the two organisms. Further to this point, a highly significant linear correlation between EC50s of both organisms was found; i.e. $EC50_{\text{rotifer}} = 0.56 \times EC50_{\text{snail}} + 3.09$ ($r = 0.99$, $p = 0.0015$, $n = 5$). This already suggests that water chemistry affects chronic Zn toxicity to both organisms in a nearly identical manner. This is explored further in the BLM section (see Section 3.4). Further, the estimated regression equation shows that the rotifer *B. calyciflorus* is on average about twice as sensitive to Zn as the snail *L. stagnalis* (see also 3.5).

3.3. Effect of Ca, pH, and DOC on chronic Zn toxicity

It has been shown that pH, Ca, and DOC are the main variables affecting chronic zinc toxicity to freshwater algae, daphnids and fish (De Schamphelaere and Janssen, 2004; De Schamphelaere et al., 2005; Heijerick et al., 2005; Wilde et al., 2005; Brinkman and Woodling, 2005; Van Sprang et al., 2009). Hence, it is instructive to check if our datasets suggest effects of these parameters on *L. stagnalis* and *B. calyciflorus* as well.

The individual effect of pH and Ca on chronic Zn toxicity to both species can easily be observed from our data, since the experimental design included the synthetic modification of the pH and Ca concentration of the natural Brisy water (see 2.2). Fig. 4 clearly shows that for both species, increases of the pH and Ca result in increased EC10s and EC50s (decreased toxicity). The increase of pH from 6.8 to 8.2 results in ECx values that are 35% to 41% higher for both species (Table 2). Investigation of the detailed speciation output of the BLM calculations (see supportive information) indicates that the effect of pH in Brisy water is likely due to increased complexation of Zn^{2+} with CO_3^{2-} into the $ZnCO_3$ complex, as its abundance increased from <1% of the dissolved Zn at low pH to 26% to 38% at high pH. Thus, the increased complexation leaves less Zn^{2+} ions to contribute to toxicity, which likely explains the reduced toxicity at high pH. Of course, since the adjustment to high pH was established through the addition of $NaHCO_3$, it cannot be entirely ruled out that the observed reduction of toxicity with increased pH is in fact partly due to increased competition between Na^+ and Zn^{2+} ions. Although this is

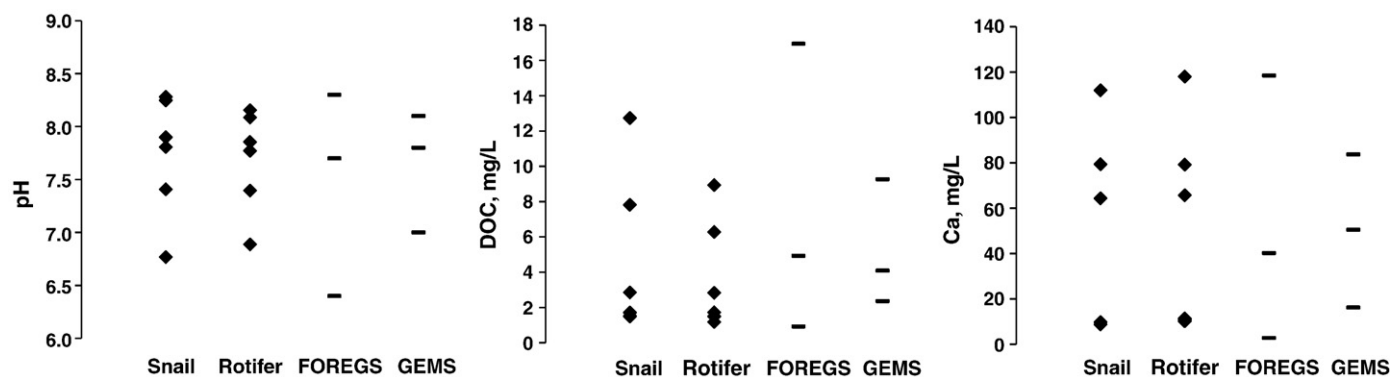


Fig. 1. Measured pH, Ca, and dissolved organic carbon (DOC) in snail and rotifer exposure media (diamonds) compared with 10th, 50th, and 90th percentiles (horizontal bars) of these variables reported in the FOREGS and GEMS freshwater monitoring databases (see Section 3.1 for details).

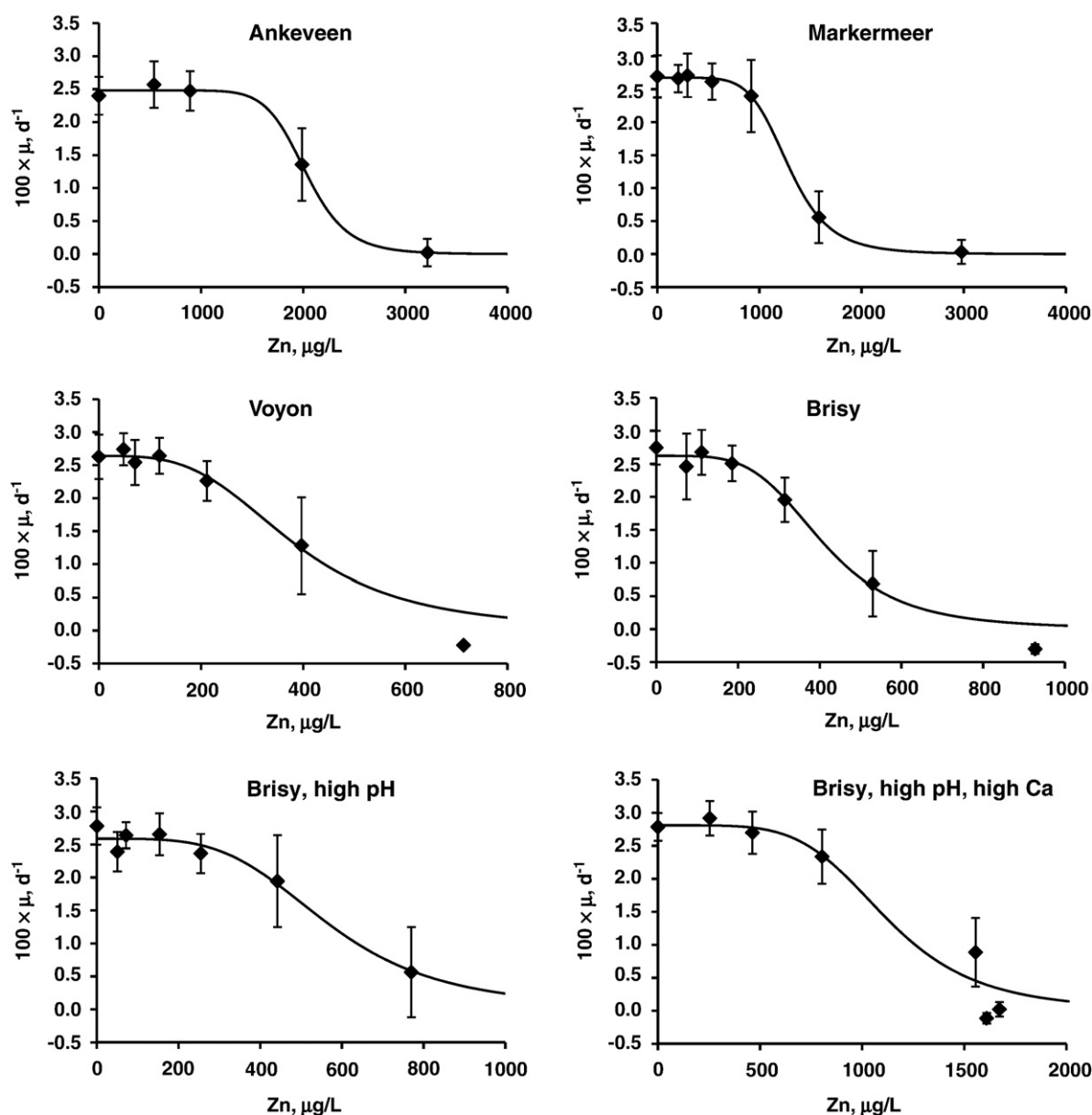


Fig. 2. Concentration response data and fitted concentration response curves for *Lymnaea stagnalis*. Error bars denote $\pm$ one standard deviation.

less likely, given the rather weak competitive effect between Na^+ and Zn^{2+} for the biotic ligand of *D. magna* (Heijerick et al., 2005), additional investigations on the individual effect of Na, e.g. through addition of $NaSO_4$, would be required to make a more definitive conclusion.

The increase of Ca from about 10 (0.25 mmol/L) to about 120 mg/L (3 mmol/L) resulted in a 2.0 to 2.6-fold increase of the EC_x values (both species, Table 2, Fig. 4). This appears to be a slightly lower protective effect as experienced by *D. magna*, for which chronic Zn toxicity was reduced by 3-fold following an increase of Ca from 0.25 to 3 mmol/L of Ca (Heijerick et al., 2005).

Finally, the individual effect of DOC on chronic Zn toxicity to *L. stagnalis* and *B. calyciflorus* is the least easy one to disentangle from the conducted toxicity tests, because the DOC concentration was not modified in a univariate way. However, if we only consider the media which are approximately similar in pH and Ca, but are different in DOC level, i.e. Ankeveen, Markermeer, and Brisy (adjusted to high pH and high Ca), a clear increase of the EC₅₀ levels with increasing DOC is observed for both species (Fig. 5). These results clearly suggest

that increasing the DOC concentration reduces chronic toxicity of Zn to *L. stagnalis* and *B. calyciflorus*.

3.4. Biotic ligand model predictions

The advantage of using the BLM to interpret and explain variation in toxicity data among different exposure media is that it considers the effects of all physico-chemical variables simultaneously, and not individually, as above (see Section 3.3). In this context, the primary aim of the present study was to investigate if the chronic Zn BLM developed for one invertebrate, i.e. *D. magna* (phylum Arthropoda) would be able to accurately predict the toxicity variation among physico-chemically different waters for two other invertebrates from different phyla, i.e. *L. stagnalis* (phylum Mollusca) and *B. calyciflorus* (phylum Rotifera). Similar to other cross-species extrapolation investigations of BLMs (Santore et al., 2002; De Schamphelaere et al., 2006; De Schamphelaere and Janssen, 2006; Deleebeeck et al., 2007; Deleebeeck et al., 2009; Van Sprang et al., 2008; Van Sprang et al., 2009; ECB, 2009; Schlegel et al., unpublished data) we therefore

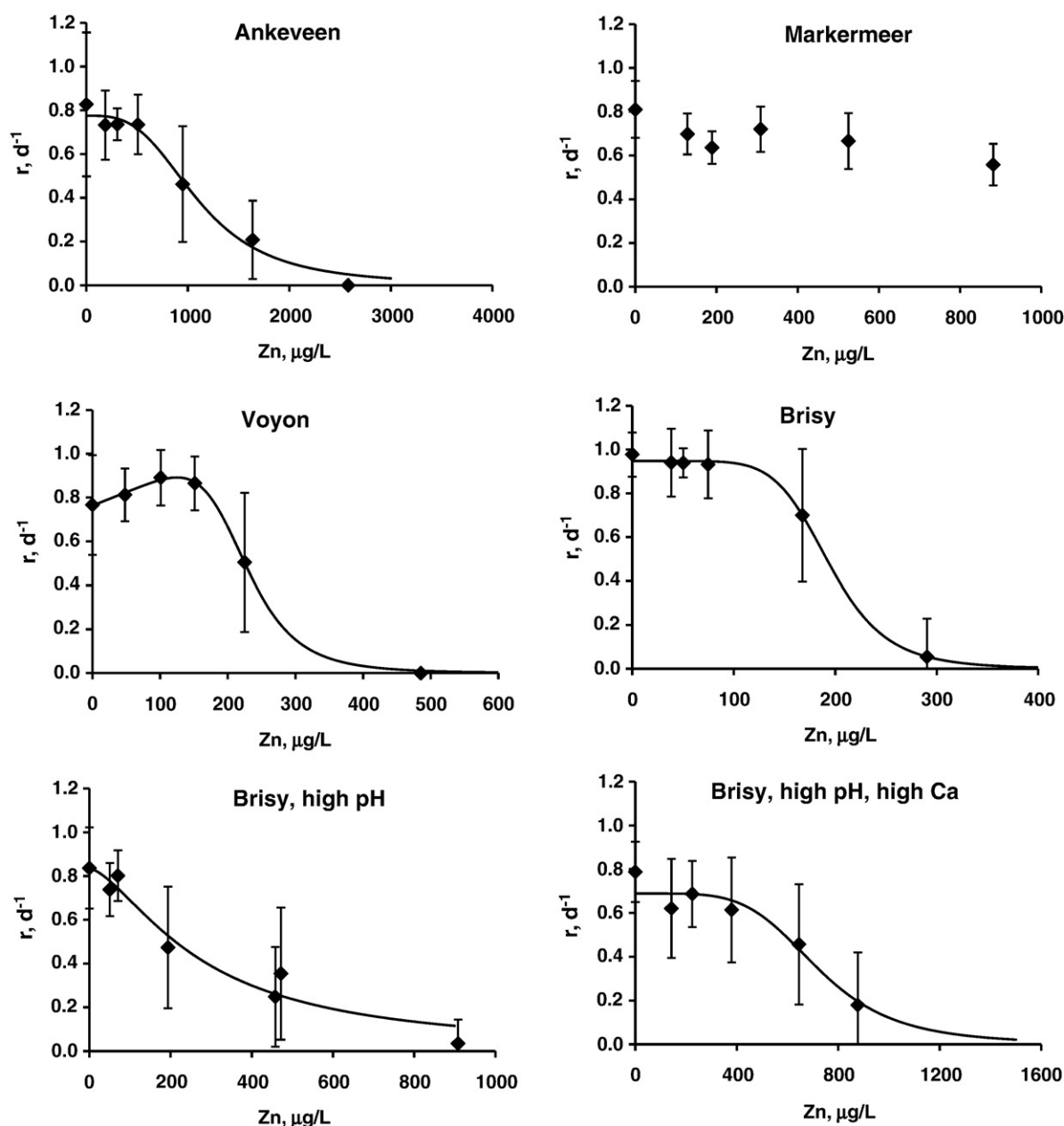


Fig. 3. Concentration response data and fitted concentration response curves for *Brachionus calyciflorus*. Error bars denote $\pm$ one standard deviation.

parameterized a chronic Zn BLM for *L. stagnalis* and *B. calyciflorus* by only optimizing the intrinsic sensitivity of these species. It was assumed that the stability constants for the competitive interactions between Ca^{2+} , Mg^{2+} , H^{+} , Na^{+} and Zn^{2+} are the same for both species as for *D. magna* (Table 1). Following the original BLM concept, the fractional occupancies of the biotic ligand sites by Zn at EC10 and EC50 levels, i.e. $f_{ZnBL,10\%}$ and $f_{ZnBL,50\%}$, are measures of the intrinsic sensitivity of a species to Zn and should be fairly constant across different media (Meyer et al., 1999; Santore et al., 2001). Thus, as a first step, $f_{ZnBL,10\%}$ and $f_{ZnBL,50\%}$ were calculated for each medium using the BLM software in 'speciation mode' (Table 3). For both species the coefficients of variation (CV, %) of $f_{ZnBL,10\%}$ and $f_{ZnBL,50\%}$ are much smaller than the EC10 and EC50 values on the basis of dissolved Zn (Table 3). This is a first clear indication that the BLM explains an important part of the variation in toxicity (expressed as dissolved Zn) among the six exposure media with different physico-chemistry.

As a next step, again similar as in other BLM cross-species extrapolation studies, the arithmetic means of all $f_{ZnBL,10\%}$ and $f_{ZnBL,50\%}$

values were selected as the final intrinsic species sensitivity parameters (Table 3). These values, together with the stability constants (assumed the same as for *D. magna*, Table 1), make up the newly parameterized chronic Zn BLMs for *L. stagnalis* and *B. calyciflorus* (Table 1). The parameter files – representing these BLMs – for use with the BLM software are given as supportive information S5 to S8 (Zn_Lymnaea_stagnalis_28d_EC10.dat, Zn_Lymnaea_stagnalis_28d_EC50.dat, Zn_Brachionus_calyciflorus_2d_EC10.dat, Zn_Brachionus_calyciflorus_2d_EC50.dat). These BLMs were then used to make predictions of ECx values (expressed as dissolved Zn) for all six waters, by running the BLM software in 'toxicity mode'. All predicted ECx values are given as supportive information (S3 and S4). These values were then compared with the observed ECx values (reported in Table 2). All predictions of EC10s and EC50s were within 1.5-fold difference from observations for *L. stagnalis*, with an average prediction error of 1.3 and 1.2-fold, respectively (Fig. 6). Predictions were only slightly less accurate for *B. calyciflorus*, with all predictions showing less than 1.6-fold error, with an average prediction error of 1.3-fold for both the EC10s and EC50s.

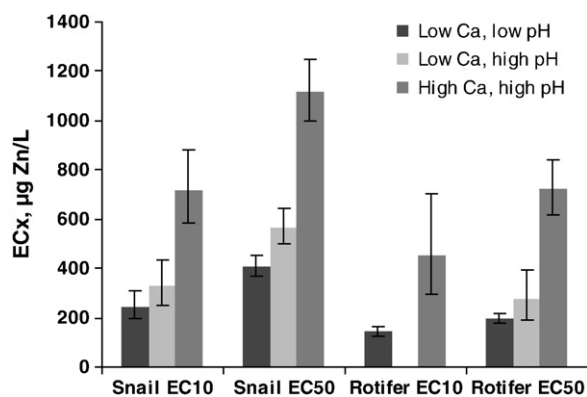


Fig. 4. ECx values obtained in Brisy water ('low Ca, low pH') and modified Brisy waters to show the effect of pH (at low Ca) and of Ca (at high pH) on chronic Zn toxicity to *L. stagnalis* (snail) and *B. calyciflorus* (rotifer). Error bars delineate 95% confidence intervals. See Table 2 for measured values of pH and Ca.

This accuracy is at least equally good as for chronic Zn BLM predictions of chronic Zn EC50s for *D. magna* itself in 6 field waters (4 of which are from the same location as in the present study), for which an average prediction error of 1.5-fold was observed (De Schamphelaere et al., 2005). Thus, the BLM developed for *D. magna* can be extrapolated with confidence to two species within different phyla of the animal kingdom. This means that the competitive interactions at the biotic ligand of three aquatic invertebrate species can be accurately described with the same set of values for $\log K_{ZnBL}$, $\log K_{CaBL}$, $\log K_{MgBL}$, $\log K_{HBL}$, and $\log K_{NaBL}$. Given that this set of constants is also fairly similar to the set of constants used to predict chronic Zn toxicity to fish species, gives further confidence that the interactions that take place between Zn^{2+} and competing cations at biological surfaces are rather well conserved across the animal kingdom, as suggested earlier (De Schamphelaere and Janssen, 2004; Van Sprang et al., 2009).

3.5. Comparison of intrinsic species sensitivities

Intrinsic sensitivities of several invertebrate species are presented in Table 1. With an $f_{ZnBL,10\%}$ of 0.121, *B. calyciflorus* is about twice as sensitive than *L. stagnalis* ($f_{ZnBL,10\%} = 0.203$). However, both species are less sensitive than 8 out of 9 other invertebrates considered in a recent review (Van Sprang et al., 2009). Based on toxicity data, published in other ecotoxicity studies, these authors calculated values of $f_{ZnBL,10\%}$ between 0.013 (for another freshwater snail, i.e. *Potamo-pyrghus jenkinsii*) and 0.194 (for the larvae of the mayfly *Ephoron virgo*). *B. calyciflorus* was 2.5-fold less sensitive than the two other

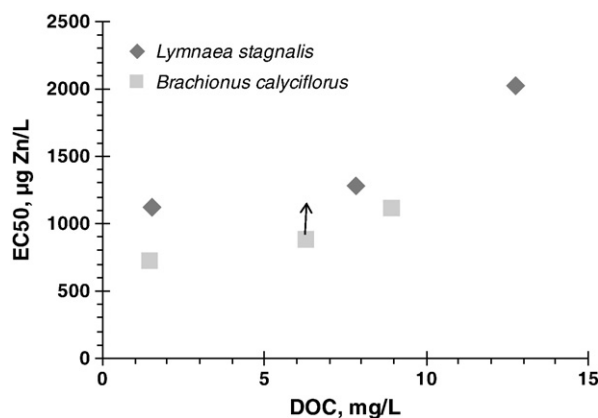


Fig. 5. The effect of dissolved organic carbon on chronic EC50s for Zn. Only data with similar pH and Ca, i.e. in the test waters Ankeveen, Markermeer and Brisy (adjusted to high pH and high Ca) are plotted. The $\uparrow$ denotes that the true EC50 is higher than this value (see footnote in Table 2).

Table 3

Mean, standard deviation and coefficient of variation (CV%) of ECx values expressed as dissolved Zn and as fraction of the biotic ligand occupied by Zn at the ECx.

Medium	Dissolved Zn (µg/L)			$f_{ZnBL, x\%}$		
	Mean	stdev	CV%	Mean	stdev	CV%
<i>Lymnaea stagnalis</i>						
EC10 (n = 6)	672	547	81	0.203	0.067	33
EC50 (n = 6)	964	642	67	0.295	0.063	21
<i>Brachionus calyciflorus</i>						
EC10 (n = 4)	335	197	59	0.121	0.035	29
EC50 (n = 5)	508	395	78	0.170	0.053	31

rotifers considered, i.e. *Brachionus rubens* and *Aenuropsis fissa*. *L. stagnalis* appears to be intrinsically the least sensitive invertebrate so far. The low chronic sensitivity of *L. stagnalis* to Zn is surprising, given that Zn is a known Ca-antagonist (Hogstrand et al., 1996; Muysen et al., 2006) and that freshwater pulmonate snails have been shown to be among the most sensitive organisms to other Ca antagonists, including Pb (Borgmann et al., 1978; Grosell et al., 2006) and Co (De Schamphelaere et al., 2008). There is some evidence that Pb and Co inhibit Ca uptake in snails, which then becomes a limiting factor for shell growth (Grosell et al., 2006; De Schamphelaere et al., 2008). The unexpected relative insensitivity of *L. stagnalis* to Zn therefore deserves further attention.

3.6. Implications for risk assessment

The present study fills an important data gap with respect to the application of the BLM-based bioavailability normalization of data on chronic Zn toxicity to freshwater species, identified by Van Sprang et al. (2009). While some evidence was already available on the possibility of extrapolation of chronic Zn BLMs across fish and algae species, no such evidence was available for the large group of invertebrate species (Van Sprang et al., 2009). The earlier lack of such knowledge has been one of the main reasons that prompted the EU regulatory authorities to adopt a rather conservative approach to deal with bioavailability of Zn in the RAR, called the "bioavailability factor approach" (BioF approach), instead of the so-called full SSD bioavailability normalization, which has been advocated as more valuable and robust method by Van Sprang et al. (2009) for Zn and which has also been implemented in the EU RAR's for Cu and Ni (Van Sprang et al., 2008; ECB, 2009) and in the US EPA ambient freshwater copper criteria calculations (USEPA, 2007). In brief, the full SSD normalization approach entails the use of the chronic Zn BLMs for bioavailability normalization of all available high quality chronic ecotoxicity data for all species to region, river basin or site-specific physico-chemistry, followed by estimating hazardous concentrations for 5% of the organisms (HC5) by fitting a species sensitivity distribution (SSD) to the normalized ecotoxicity dataset (Van Sprang et al., 2009). Ecotoxicity data with fish, algae, and invertebrates are normalized using chronic Zn BLMs developed for *O. mykiss*, *P. subcapitata*, and *D. magna*, respectively, with the crucial assumption that chronic Zn BLMs can be extrapolated with confidence across different species. The present study gives confidence to this assumption with respect to using the *D. magna* based Zn BLM for normalizing existing chronic toxicity for other invertebrate species obtained in a medium with known physico-chemistry, to the physico-chemical conditions of any given water for which a risk assessment needs to be performed. Combined with BLM-normalized toxicity data for fish and algae, these data can be entered in a species sensitivity distribution (SSD) analysis to derive bioavailability normalized HC5 values. Given that this approach was followed in the river basin specific risk assessments of Zn by Van Sprang et al. (2009), our study gives additional support to the conclusions drawn in that study. Further, it gives confidence to using this approach in other regulatory

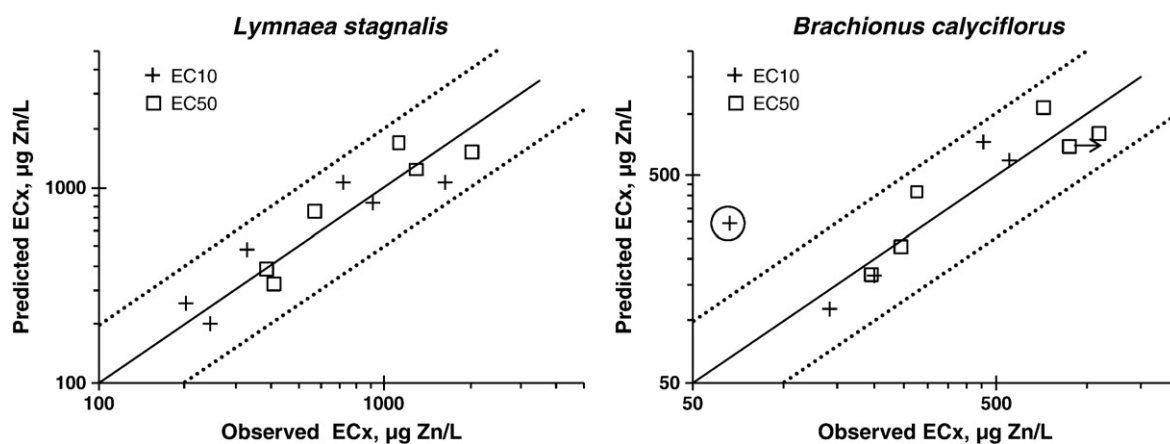


Fig. 6. Observed and Biotic Ligand Model predicted 10% (EC10) and 50% effective concentrations (EC50) of dissolved Zn for two species. The full line indicates a 1:1 ratio between observation and prediction; the dotted line indicates the range of two-fold difference between observation and prediction. For *B. calyciflorus*, the encircled EC10 is considered unreliable (see text), while the true EC50 of the square marked with an arrow ($\rightarrow$) is higher than the depicted square (see footnote in Table 2).

actions worldwide, including for example REACH and the derivation of environmental quality standards (EQS) in the context of the Water Framework Directive (2000/60/EC).

As an example, we have applied this full SSD normalization approach to a typical water composition of an EU river basin, i.e. the one described as Danube sample #5 by Van Sprang et al. (2009), with pH 7.95, DOC = 2.6 mg/L, and Ca = 45.9 mg/L (Fig. 7). Since the two species investigated in the present study are among the least sensitive from the entire dataset of 21 species in total, their addition to the SSD causes a slight upward shift of the HC5. By fitting the lognormal distribution, a theoretical HC5 of 24.5 µg Zn/L was found, while without the addition of the snail and rotifer data from the present study, the HC5 was 21.6 µg Zn/L (14% lower). Thus, the addition to the SSD of the chronic toxicity data for the two species in the present study, will likely slightly increase HC5 estimates and reduce risk estimates, compared to our previous analysis (Van Sprang et al., 2009).

4. Conclusion

We investigated if the chronic zinc biotic ligand model (BLM) developed earlier for the arthropod *D. magna* could be extrapolated to predict chronic ecotoxicity of zinc as a function of water chemistry to two species from other phyla, i.e. the mollusc *L. stagnalis* and the rotifer

B. calyciflorus. We chronically exposed these two species to zinc in six natural surface waters and found an up to 8.1-fold variation in toxicity among these waters, with evidence that higher pH and higher concentrations of Ca and DOC were associated with lower toxicity (higher ECx values). The chronic Zn BLM for *D. magna*, when calibrated only to reflect the intrinsic sensitivity of *L. stagnalis* and *B. calyciflorus*, was able to predict all ECx values with a less than 1.6-fold error, which demonstrates that the chronic *D. magna* Zn BLM can be extrapolated to other invertebrate phyla. This lends further support to the use of the chronic Zn BLM to account for bioavailability of zinc – through the full SSD bioavailability normalization approach – in risk assessment and also in the derivation of environmental quality standards.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [10.1016/j.scitotenv.2010.07.043](https://doi.org/10.1016/j.scitotenv.2010.07.043).

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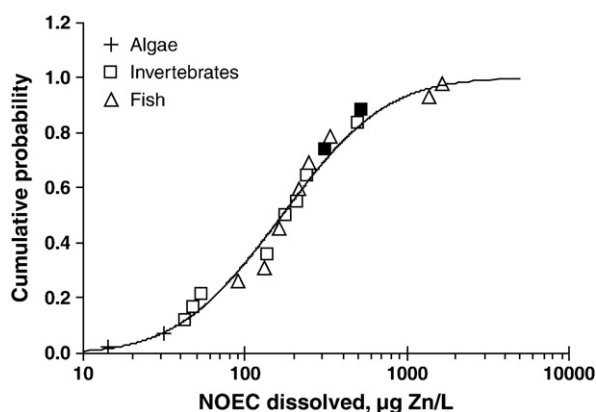


Fig. 7. Empirical cumulative probability distribution (crosses, squares and triangles) of chronic no observed effect concentrations (or EC10s) normalized to typical conditions of a EU river basins (see text for details). Open symbols represent toxicity data from species that were already present in the ecotoxicity database (Van Sprang et al., 2009). Filled symbols represent the data for *L. stagnalis* and *B. calyciflorus* from the present study. The full line represents the lognormal distribution fitted to the NOEC values.

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